

4.10.3 Using materials (chemistry only)

4.10.3.1 Corrosion and its prevention

Content	Key opportunities for skills development
Corrosion is the destruction of materials by chemical reactions with substances in the environment. Rusting is an example of corrosion. Both air and water are necessary for iron to rust. Corrosion can be prevented by applying a coating that acts as a barrier, such as greasing, painting or electroplating. Aluminium has an oxide coating that protects the metal from further corrosion. Some coatings are reactive and contain a more reactive metal to provide sacrificial protection, eg zinc is used to galvanise iron.	
Students should be able to: • describe experiments and interpret results to show that both air and water are necessary for rusting • explain sacrificial protection in terms of relative reactivity.	WS 2.2, 7, 3.5 Investigate the conditions for rusting.

4.10.3.2 Alloys as useful materials

Content	Key opportunities for skills development
Most metals in everyday use are alloys. Bronze is an alloy of copper and tin. Brass is an alloy of copper and zinc. Gold used as jewellery is usually an alloy with silver, copper and zinc. The proportion of gold in the alloy is measured in carats. 24 carat being 100% (pure gold), and 18 carat being 75% gold. Steels are alloys of iron that contain specific amounts of carbon and other metals. High carbon steel is strong but brittle. Low carbon steel is softer and more easily shaped. Steels containing chromium and nickel (stainless steels) are hard and resistant to corrosion. Aluminium alloys are low density.	
Students should be able to: • recall a use of each of the alloys specified • interpret and evaluate the composition and uses of alloys other than those specified given appropriate information.	MS 1a Recognise and use expressions in decimal form. MS 1c Use ratios, fractions and percentages.

4.10.3.3 Ceramics, polymers and composites

Content	Key opportunities for skills development
Most of the glass we use is soda-lime glass, made by heating a mixture of sand, sodium carbonate and limestone. Borosilicate glass, made from sand and boron trioxide, melts at higher temperatures than soda-lime glass. Clay ceramics, including pottery and bricks, are made by shaping wet clay and then heating in a furnace. The properties of polymers depend on what monomers they are made from and the conditions under which they are made. For example, low density (LD) and high density (HD) poly(ethene) are produced from ethene. Thermosoftening polymers melt when they are heated. Thermosetting polymers do not melt when they are heated. Students should be able to: • explain how low density and high density poly(ethene) are both produced from ethene • explain the difference between thermosoftening and thermosetting polymers in terms of their structures. Most composites are made of two materials, a matrix or binder surrounding and binding together fibres or fragments of the other material, which is called the reinforcement. Students should be able to recall some examples of composites.	
Students should be able to, given appropriate information: • compare quantitatively the physical properties of glass and clay ceramics, polymers, composites and metals • explain how the properties of materials are related to their uses and select appropriate materials.	WS 1.4, 3.5, 3.8 Compare the properties of thermosetting and thermosoftening polymers.